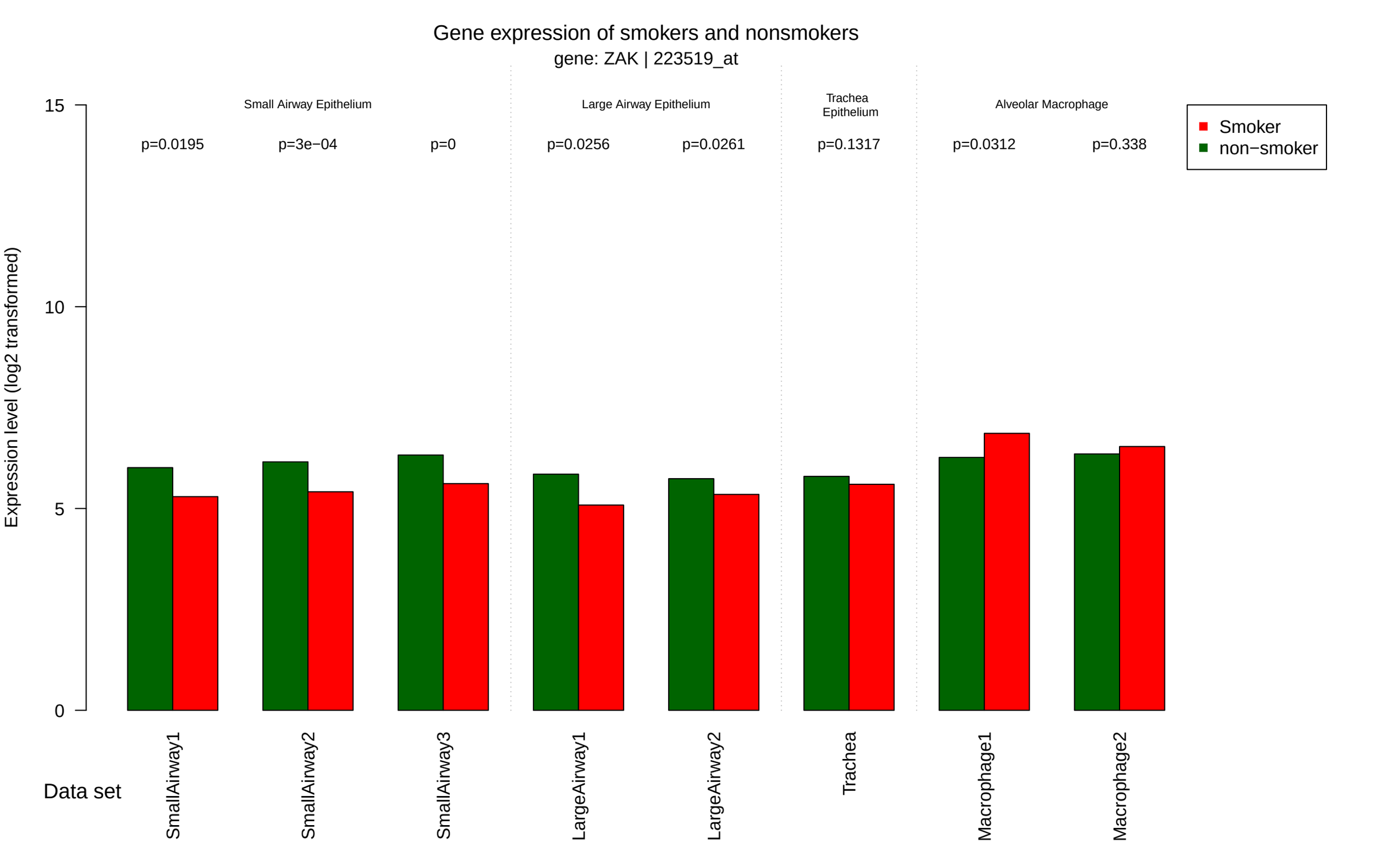
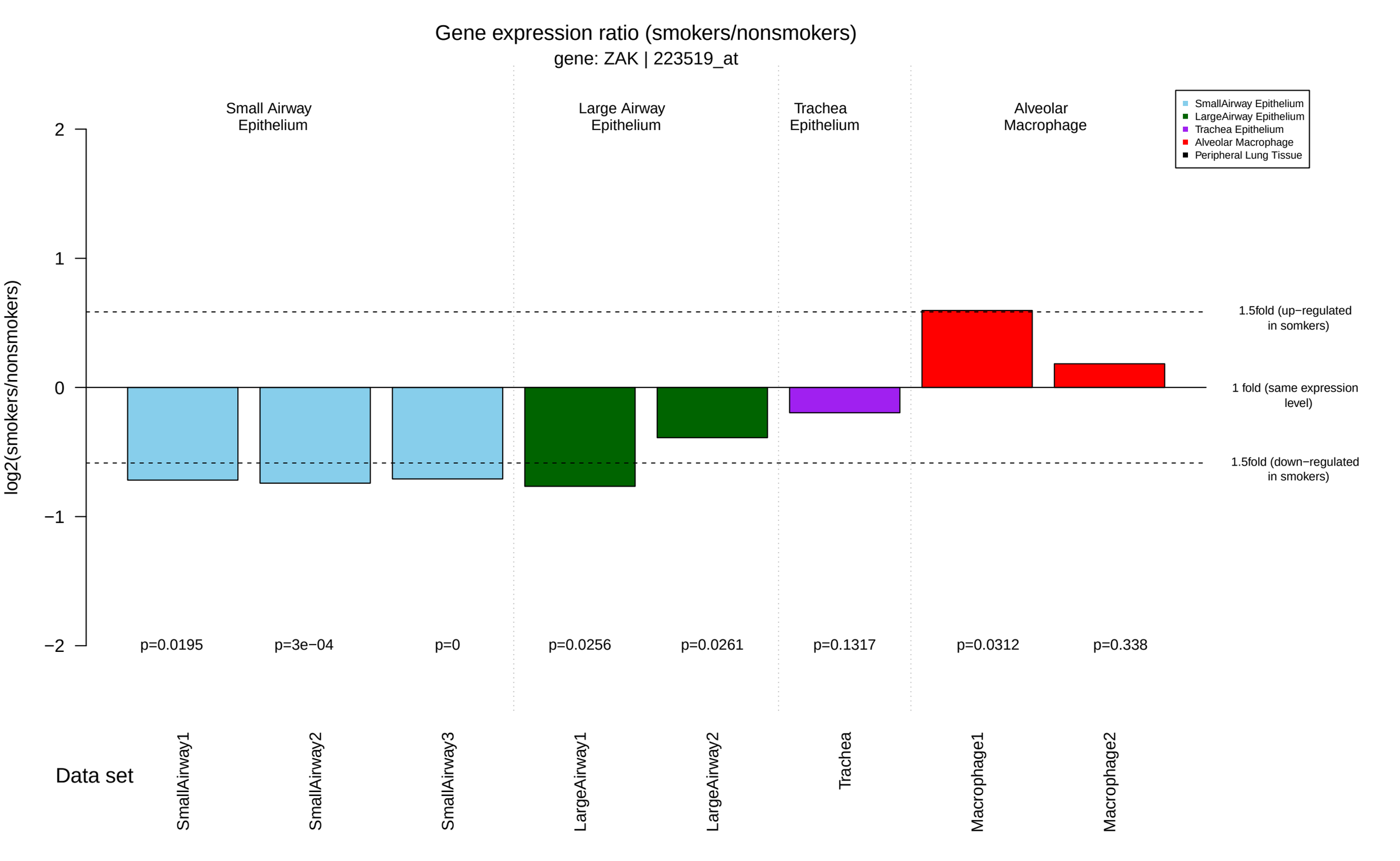
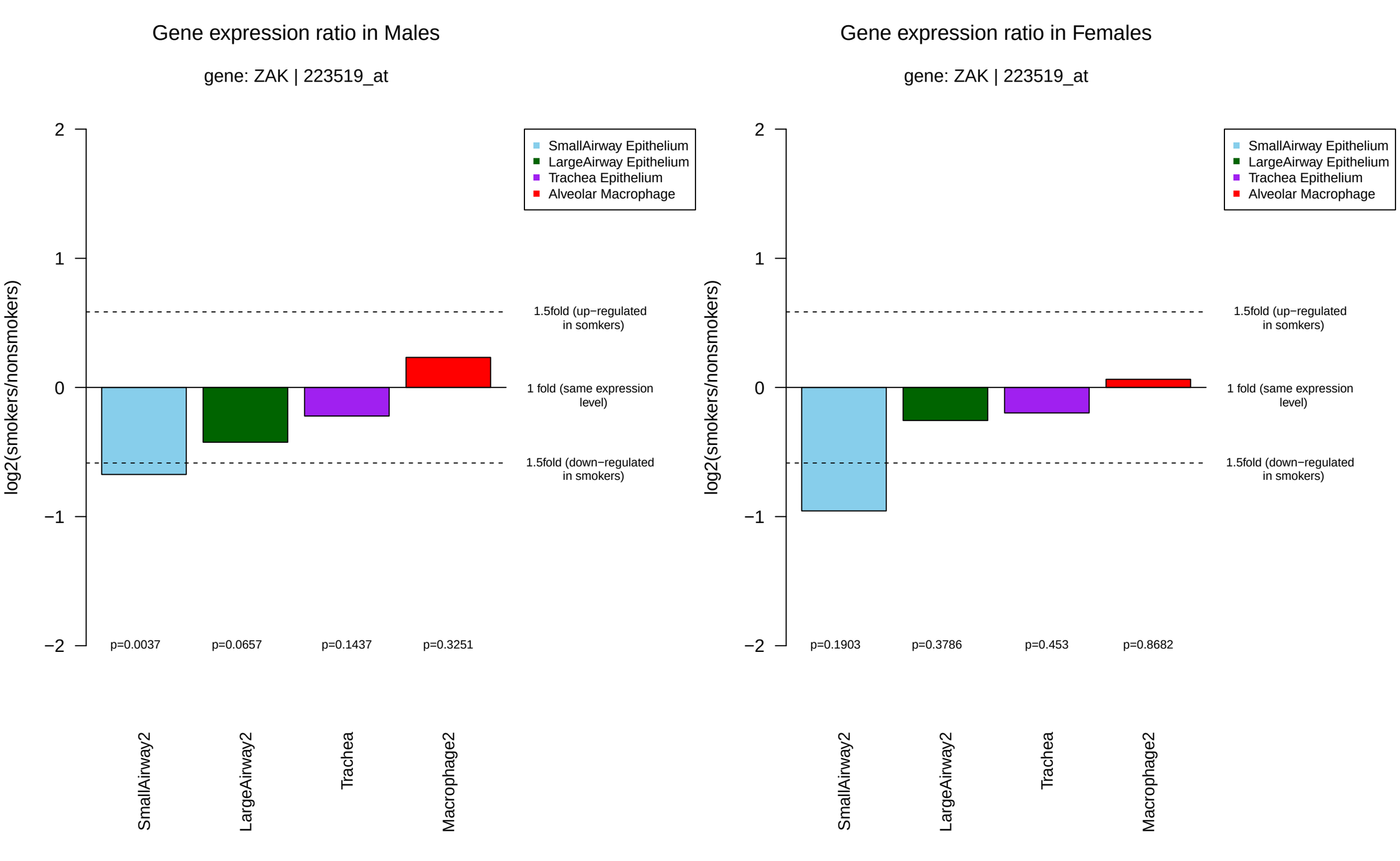
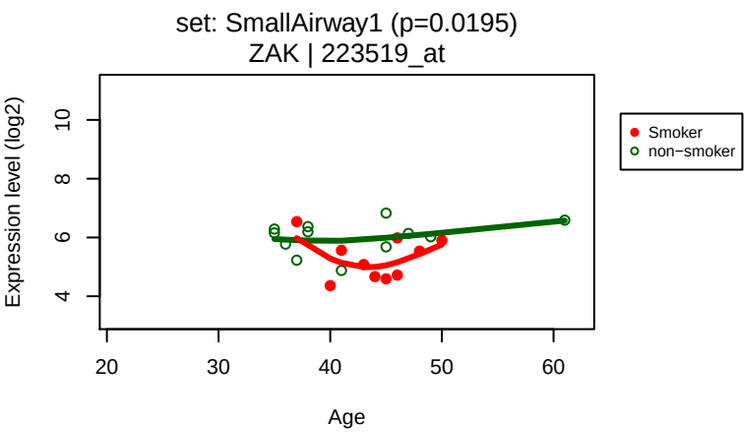
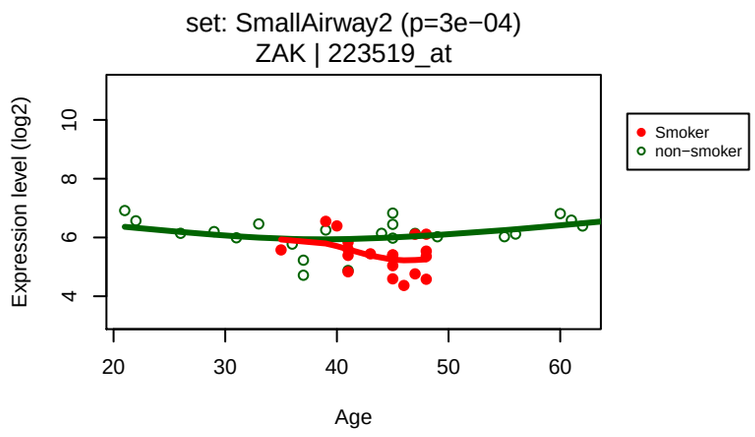
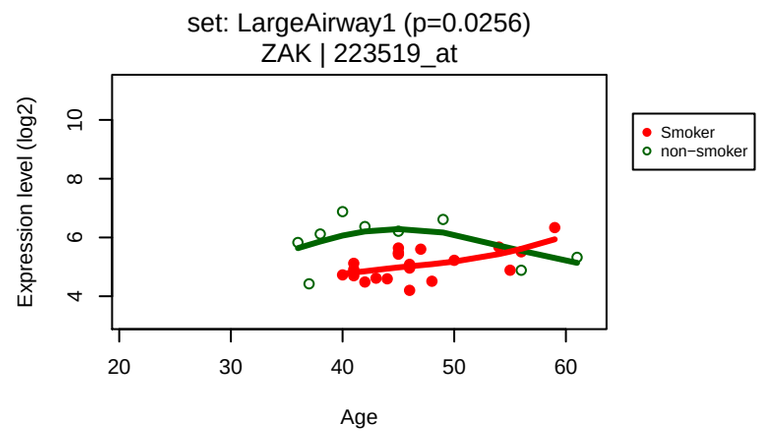

Gene expression of smokers and nonsmokers
gene: ZAK | 223519_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0195
p=3e−04
p=0
p=0.0256
p=0.0261
p=0.1317
p=0.0312
p=0.338
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

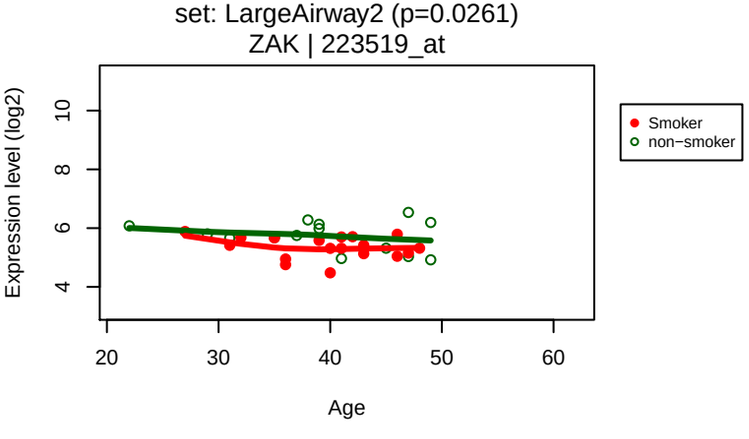

Gene expression ratio (smokers/nonsmokers)
gene: ZAK | 223519_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0195
p=3e−04
p=0
p=0.0256
p=0.0261
p=0.1317
p=0.0312
p=0.338
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

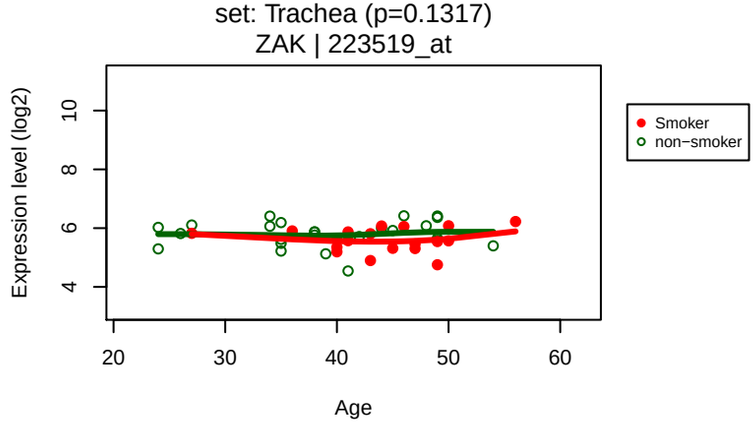

Gene expression ratio in Males
gene: ZAK | 223519_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.0657
p=0.1437
p=0.3251
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZAK | 223519_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1903
p=0.3786
p=0.453
p=0.8682
SmallAirway2
LargeAirway2
Trachea
Macrophage2

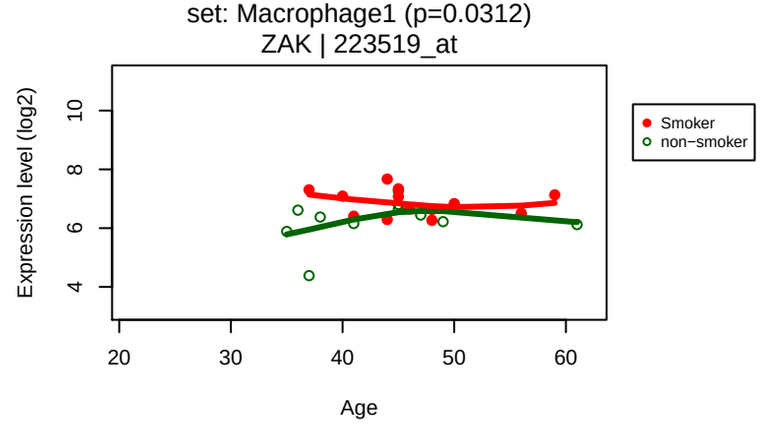

set: SmallAirway1 (p=0.0195)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker

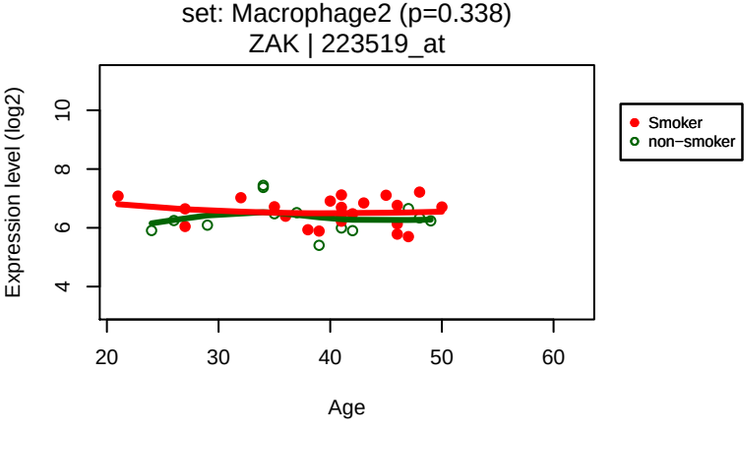

set: SmallAirway2 (p=3e−04)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker

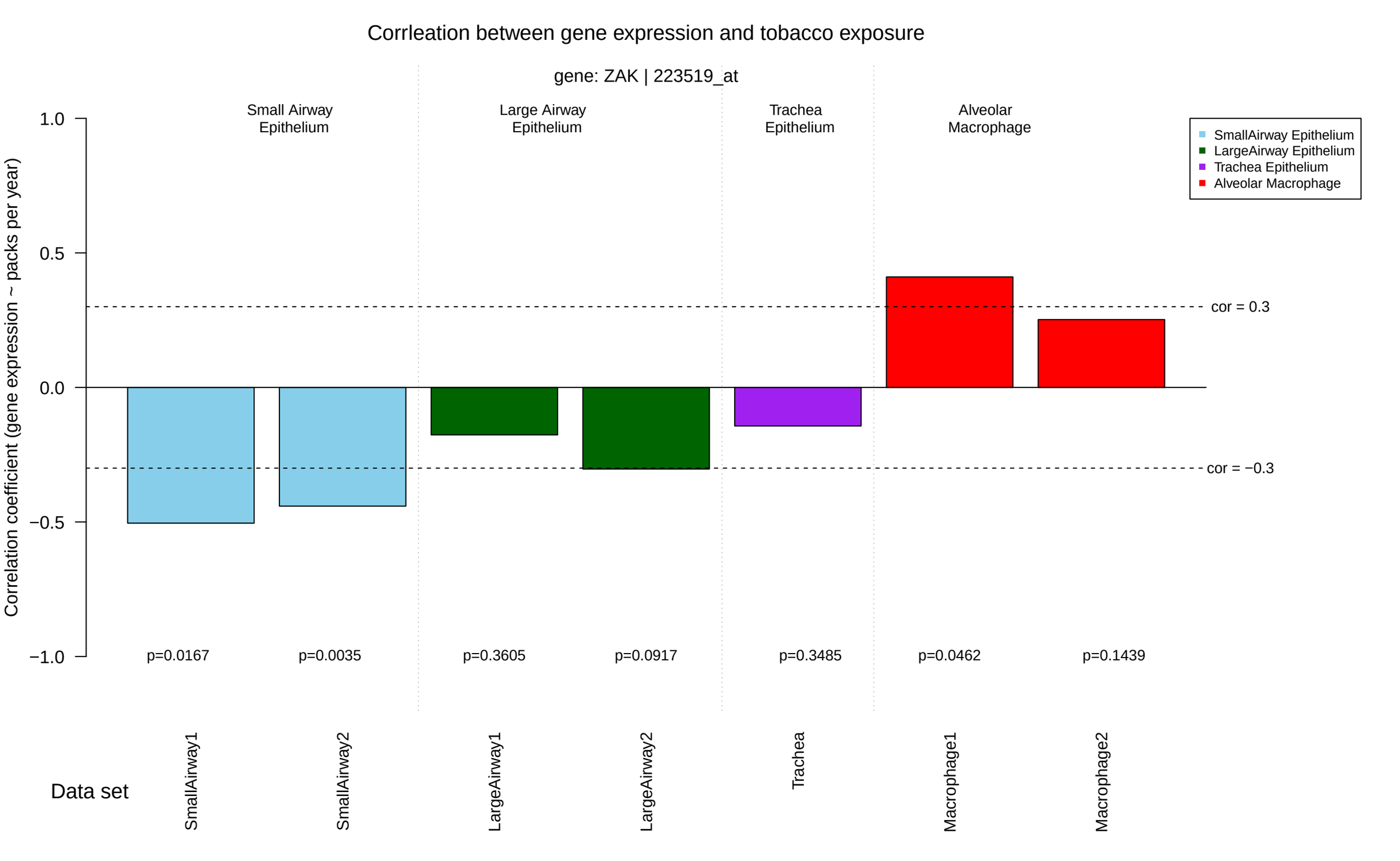

set: LargeAirway1 (p=0.0256)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0261)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1317)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0312)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.338)
ZAK | 223519_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZAK | 223519_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0167
p=0.0035
p=0.3605
p=0.0917
p=0.3485
p=0.0462
p=0.1439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2